

Simplex III: Two-Phase Simplex

How can simplex find a feasible starting basis, then solve the original problem?

3.5 · Finding an initial feasible basis

Lecture 7 · Bertsimas–Tsitsiklis §3.5

Use the pivot, reduced-cost, and anticycling rules developed in Simplex I and II.

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How Do We Find the First Feasible Basis?

An anticycling rule tells simplex how to continue without repeating bases. We still need a place to start.

The missing input

A **basic feasible solution**, together with its basis matrix and constraint rows.

The two-phase idea

Phase I: find a feasible basis or prove infeasibility.

Phase II: optimize the original objective from that basis.

An Easy Start: Nonnegative Slacks

Suppose A has m rows and n columns, and the constraints are

$$Ax \leq b, \quad x \geq 0, \quad b \geq 0.$$

Add a vector $s \in \mathbb{R}^m$ of nonnegative **slack variables**:

$$Ax + s = b, \quad x, s \geq 0.$$

$$x = 0, \quad s = b, \quad B = I_m.$$

The slack columns form the identity matrix I_m ; their values are nonnegative. This is a basic feasible solution.

The Usual Starting Dictionary Can Be Infeasible

$$x_1 + x_2 \geq 2, \quad x_1 + 2x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

Subtract **surplus** $s_1 \geq 0$.

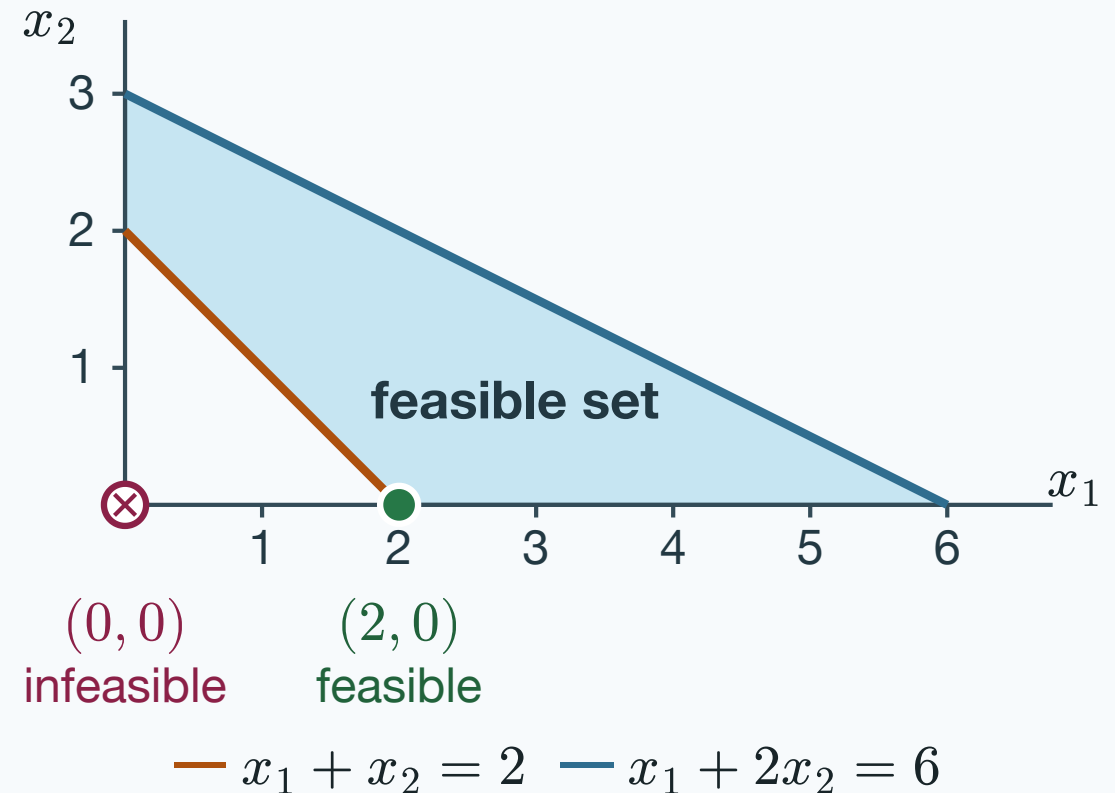
Add **slack** $s_2 \geq 0$:

$$s_1 = -2 + x_1 + x_2,$$

$$s_2 = 6 - x_1 - 2x_2.$$

Set $x_1 = x_2 = 0$: then $s_1 = -2$, $s_2 = 6$. The starting dictionary is **infeasible**.

Yet $(x_1, x_2) = (2, 0)$ satisfies both constraints.
How can we find a feasible basis **systematically**?



An Artificial Variable Gives a Feasible Start

Keep s_2 basic. Add an **artificial variable** $y \geq 0$ to the first equality:

$$x_1 + x_2 - s_1 + y = 2.$$

Solve for the new basic variables y, s_2 :

$$\begin{aligned} y &= 2 - x_1 - x_2 + s_1, \\ s_2 &= 6 - x_1 - 2x_2. \end{aligned}$$

Set the nonbasic variables $x_1 = x_2 = s_1 = 0$. Then $y = 2, s_2 = 6$:
we have a **feasible auxiliary dictionary**. All five variables are nonnegative.

Phase I: minimize $w = y$ subject to this dictionary and nonnegativity.

When $y = 0$, we recover the original constraints. Keep the legitimate slack and surplus variables.

The General Phase I Problem

Write the original LP as $\min c^\top x$ subject to $Ax = b$, $x \geq 0$. Here x includes any slack and surplus variables.

For m equalities, a general construction uses **one artificial variable per row**:

$$\begin{aligned} \min \quad & w = \sum_{i=1}^m y_i \\ \text{subject to} \quad & Ax + y = b, \\ & x \geq 0, \quad y \geq 0. \end{aligned}$$

Intended start: $x = 0$, $y = b$, $B = I_m$.

Make this start feasible: first make $b \geq 0$. If $b_i < 0$, multiply **that entire equality** by -1 **before adding the helpers**. Then $y = b \geq 0$.

Since $y \geq 0$, total help $w = 0$ means **every helper is zero**. The original cost $c^\top x$ waits until Phase II.

Why Zero Phase I Cost Means Feasibility

If $Ax = b, x \geq 0$, then $(x, 0)$ satisfies the auxiliary constraints and has $w = 0$.

Conversely, if an auxiliary feasible point has $w = 0$,

$$y \geq 0, \quad \sum_i y_i = 0 \quad \implies \quad y = 0.$$

Thus $Ax + y = b$ becomes $Ax = b$.

Original LP feasible $\iff w^* = 0.$

The symbol w^* denotes the optimal Phase I objective value. Nonnegativity of each artificial variable is essential.

A Positive Phase I Optimum Proves Infeasibility

Phase I starts feasible and its objective satisfies $w \geq 0$, so it cannot be unbounded below.

$$w^* > 0 \implies \text{the original LP is infeasible.}$$

An original feasible point would give auxiliary cost zero, contradicting optimality of $w^* > 0$.

A **current** positive Phase I cost is inconclusive. Continue until the auxiliary optimum is certified. A feasible auxiliary point of cost zero already reaches the lower bound.

Example 3.8: Start Without an Obvious Basis

We will carry this complete model through both phases:

$$\begin{array}{ll}\min & f = x_1 + x_2 + x_3 \\ \text{subject to} & x_1 + 2x_2 + 3x_3 = 3, \\ & x_1 - 2x_2 - 6x_3 = -2, \\ & 4x_2 + 9x_3 = 5, \\ & 3x_3 + x_4 = 1, \quad x_1, x_2, x_3, x_4 \geq 0.\end{array}$$

Reverse the second equality to make its right-hand side positive. Keep the original costs $c = (1, 1, 1, 0)$.

Example 3.8: Build the Auxiliary LP

In this example, name the four artificial variables x_5, x_6, x_7, x_8 .

$$\begin{array}{ll}\min & w = x_5 + x_6 + x_7 + x_8 \\ \text{subject to} & x_1 + 2x_2 + 3x_3 + x_5 = 3, \\ & -x_1 + 2x_2 + 6x_3 + x_6 = 2, \\ & 4x_2 + 9x_3 + x_7 = 5, \\ & 3x_3 + x_4 + x_8 = 1, \quad x_1, \dots, x_8 \geq 0.\end{array}$$

Original variables $x_1, \dots, x_4$ have zero Phase I cost; every artificial variable has cost one.

Example 3.8: Solve for the Basic Variables

Choose x_5, x_6, x_7, x_8 as basic. Move the other terms to the right in each equality:

$$x_5 = 3 - x_1 - 2x_2 - 3x_3,$$

$$x_6 = 2 + x_1 - 2x_2 - 6x_3,$$

$$x_7 = 5 - 4x_2 - 9x_3,$$

$$x_8 = 1 - 3x_3 - x_4.$$

Set the nonbasic variables $x_1 = x_2 = x_3 = x_4 = 0$:

$$(x_5, x_6, x_7, x_8) = (3, 2, 5, 1).$$

Every value is nonnegative, so this is a **feasible starting dictionary**.

Its Phase I cost is $w = 3 + 2 + 5 + 1 = 11$. Next, express w using only the nonbasic variables.

Substitute the Dictionary into the Phase I Objective

Start with $w = x_5 + x_6 + x_7 + x_8$. Substitute the four expressions we just obtained:

$$\begin{aligned} w &= (3 - x_1 - 2x_2 - 3x_3) \\ &\quad + (2 + x_1 - 2x_2 - 6x_3) \\ &\quad + (5 - 4x_2 - 9x_3) \\ &\quad + (1 - 3x_3 - x_4). \end{aligned}$$

The x_1 terms cancel. Collect the remaining terms:

$$w = 11 - 8x_2 - 21x_3 - x_4.$$

The coefficient $\bar{q}_j$ of x_j is its **Phase I reduced cost**. For x_1, x_2, x_3, x_4 , these are 0, -8 , -21 , -1 .

From the Dictionary to the Initial Tableau

Move nonbasic terms left in each constraint, e.g. $x_1 + 2x_2 + 3x_3 + x_5 = 3$.

Use $-w$ for the objective row: $-w - 8x_2 - 21x_3 - x_4 = -11$.

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	-11	0	-8	-21	-1	0	0	0	0
x_5	3	1	2	3	0	1	0	0	0
x_6	2	-1	2	6	0	0	1	0	0
x_7	5	0	4	9	0	0	0	1	0
x_8	1	0	0	3	1	0	0	0	1

In R_0 , $-w$ is implicit. The **RHS** (right-hand side) is $-w_0 = -11$; remaining entries are reduced costs.

The Same Calculation in Matrix Notation

Here A, b describe the original four equalities **after the sign change**. Group the variables as

$$x_{\mathcal{B}} = (x_5, x_6, x_7, x_8)^\top, \quad x_{\mathcal{N}} = (x_1, x_2, x_3, x_4)^\top.$$

$$Ax_{\mathcal{N}} + x_{\mathcal{B}} = b, \quad B = I_4 \quad \Longrightarrow \quad x_{\mathcal{B}} = b - Ax_{\mathcal{N}}.$$

This is the same four-equation dictionary.

Multiplying by $\mathbf{1}^\top = (1, 1, 1, 1)$ **adds those four equations**:

$$w = \mathbf{1}^\top x_{\mathcal{B}} = \mathbf{1}^\top b - \mathbf{1}^\top Ax_{\mathcal{N}}.$$

$$\mathbf{1}^\top b = 11, \quad \mathbf{1}^\top A = (0, 8, 21, 1).$$

Thus $w = 11 - 8x_2 - 21x_3 - x_4$, exactly as obtained by substitution.

Pivot 1: x_4 Enters and x_8 Leaves

We illustrate one valid pivot sequence: choose a nonbasic variable with negative Phase I reduced cost. Other choices may also be valid. Break minimum-ratio ties by smaller basic-variable index.

Choose x_4 , whose reduced cost is -1 . Increasing $x_4 = \theta$ changes only x_8 :

$$x_8 = 1 - \theta \geq 0 \quad \implies \quad \theta^* = 1.$$

Replace basic x_8 by x_4 . Their columns are identical, so the basis matrix remains I_4 .

$$w_{\text{new}} = 11 + (-1)(1) = 10.$$

Only the objective row changes numerically:

$$R_0 \leftarrow R_0 + R_4.$$

The fourth basic-variable label changes from x_8 to x_4 .

After Pivot 1: The Cost Is 10

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−10	0	−8	−18	0	0	0	0	1
x_5	3	1	2	3	0	1	0	0	0
x_6	2	−1	2	6	0	0	1	0	0
x_7	5	0	4	9	0	0	0	1	0
x_4	1	0	0	3	1	0	0	0	1

The basis is (x_5, x_6, x_7, x_4) , with values $(3, 2, 5, 1)$.

Next choose x_3 , whose reduced cost is $−18$. Its tableau column is $(3, 6, 9, 3)^\top$.

Pivot 2: A Tie Sends x_4 Out of the Basis

For entering x_3 , divide each basic value by its positive entering-column coefficient:

Basic variable	Value	Coefficient	Step bound
x_5	3	3	1
x_6	2	6	1/3
x_7	5	9	5/9
x_4	1	3	1/3

The minimum is $\theta^* = 1/3$. The tie is between x_4 and x_6 ; choose x_4 .

$$w_{\text{new}} = 10 - 18(1/3) = 4.$$

After Pivot 2: A Basic Artificial Variable Is Zero

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	−4	0	−8	0	6	0	0	0	7
x_5	2	1	2	0	−1	1	0	0	−1
x_6	0	−1	2	0	−2	0	1	0	−2
x_7	2	0	4	0	−3	0	0	1	−3
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The basis is (x_5, x_6, x_7, x_3) . Artificial variable x_6 is still basic, but its value is now zero.

The remaining negative reduced cost is $\bar{q}_2 = -8$, so choose x_2 next.

Pivot 3: The Step Is Zero

The entering x_2 column is $(2, 2, 4, 0)^\top$, in basic order (x_5, x_6, x_7, x_3) .

$$\theta^* = \min \left\{ \frac{2}{2}, \frac{0}{2}, \frac{2}{4} \right\} = 0.$$

Basic x_6 leaves and x_2 enters at zero. The basis changes while the point and objective stay fixed:

$$w_{\text{new}} = 4 + (-8)(0) = 4.$$

Normalize the pivot row by dividing by 2; eliminate x_2 from every other row, including R_0 .

After Pivot 3: Same Point, Different Basis

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	-4	-4	0	0	-2	0	4	0	-1
x_5	2	2	0	0	1	1	-1	0	1
x_2	0	$-\frac{1}{2}$	1	0	-1	0	$\frac{1}{2}$	0	-1
x_7	2	2	0	0	1	0	-2	1	1
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The basis is (x_5, x_2, x_7, x_3) , with values $(2, 0, 2, 1/3)$. The objective remains $w = 4$.

Three nonbasic variables have negative reduced costs: x_1, x_4 , and artificial x_8 . Compare them before choosing.

Before Pivot 4: Compare the Entering Candidates

The current objective is $w = 4 - 4x_1 - 2x_4 + 4x_6 - x_8$.

Enter	Reduced cost	Maximum step	New w
x_1	-4	1	0
x_4	-2	1	2
x_8 (artificial)	-1	1	3

We choose x_1 : one pivot reaches $w = 0$. Entering x_4 is also valid.

Retain artificial columns (as here): x_8 may reenter. Increasing it decreases other artificial variables by more, so their *sum* falls.

Optional simplification: once an artificial variable leaves the basis, fix it at zero and delete its column. It then cannot reenter, even with a negative reduced cost. Every original feasible solution already has all artificial variables zero, so none is lost.

Pivot 4: Remove Another Artificial Variable

Increase x_1 , whose reduced cost is -4 . Its column is $(2, -1/2, 2, 0)^\top$, in basic order (x_5, x_2, x_7, x_3) .

$$\theta^* = \min \left\{ \frac{x_5}{2}, \frac{x_7}{2} \right\} = \min\{1, 1\} = 1.$$

The $-1/2$ coefficient makes basic x_2 increase; the zero coefficient leaves x_3 unchanged.

The leaving tie is between x_5 and x_7 . Choose x_5 , then pivot on the coefficient 2.

$$w_{\text{new}} = 4 - 4(1) = 0.$$

Phase I Reaches Zero, with x_7 Still Basic

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	0	0	0	0	0	2	2	0	1
x_1	1	1	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$-\frac{3}{4}$
x_7	0	0	0	0	0	-1	-1	1	0
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

All artificial values are zero, so the original point is $(x_1, x_2, x_3, x_4) = (1, 1/2, 1/3, 0)$.

The basis still includes artificial $x_7 = 0$. Inspect its row before starting Phase II.

Phase I Has Found a Feasible Point

In Example 3.8, $w = 0$. Every artificial variable is zero, and

$$(x_1, x_2, x_3, x_4) = (1, \frac{1}{2}, \frac{1}{3}, 0)$$

satisfies the original constraints. **Phase I has succeeded.**

The current basic variables are (x_1, x_2, x_7, x_3) , with values $(1, \frac{1}{2}, 0, \frac{1}{3})$.

Basic means solved for in the dictionary; it does not mean positive. Thus artificial x_7 is still basic at zero —a degenerate basis.

Next, write a dictionary using only original variables and restore $f = x_1 + x_2 + x_3$. We do not need to reduce w further.

Example 3.8: Read the Final Phase I Dictionary

The preceding tableau represents these equations. Nonbasic variables are x_4, x_5, x_6, x_8 :

$$\begin{aligned}x_1 &= 1 - \frac{1}{2}x_4 - \frac{1}{2}x_5 + \frac{1}{2}x_6 - \frac{1}{2}x_8, \\x_2 &= \frac{1}{2} + \frac{3}{4}x_4 - \frac{1}{4}x_5 - \frac{1}{4}x_6 + \frac{3}{4}x_8, \\x_7 &= x_5 + x_6, \\x_3 &= \frac{1}{3} - \frac{1}{3}x_4 - \frac{1}{3}x_8, \\w &= 2x_5 + 2x_6 + x_8.\end{aligned}$$

To return to the original LP, fix **all artificial variables** x_5, x_6, x_7, x_8 at zero. What does each equation become?

Set the Artificial Variables to Zero

Substitute $x_5 = x_6 = x_7 = x_8 = 0$ into the same dictionary:

$$\begin{aligned}x_1 &= 1 - \frac{1}{2}x_4, \\x_2 &= \frac{1}{2} + \frac{3}{4}x_4, \\0 &= 0, \\x_3 &= \frac{1}{3} - \frac{1}{3}x_4.\end{aligned}$$

The $x_7 = x_5 + x_6$ row becomes $0 = 0$. It imposes no restriction, so remove it. **No additional pivot is needed in this example.**

The remaining basic variables are x_1, x_2, x_3 . Setting nonbasic $x_4 = 0$ gives the same feasible point $(1, \frac{1}{2}, \frac{1}{3}, 0)$.

Remove the Redundant Tableau Row

Basic	RHS	Original variables				Artificial variables			
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
R_0	0	0	0	0	0	2	2	0	1
x_1	1	1	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$	0	0	0	$\frac{1}{3}$

The remaining basic variables are x_1, x_2, x_3 . Their nonnegative values are unchanged.

No artificial variable is now basic. We can delete all four artificial columns, including nonbasic x_7 .

Remove Artificial Columns, Keep the Constraint Rows

Basic	RHS	x_1	x_2	x_3	x_4
R_0	0	0	0	0	0
x_1	1	1	0	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$

The zero objective row still belongs to **Phase I**. It says nothing about optimality for $f = x_1 + x_2 + x_3$.

Use the original costs $c = (1, 1, 1, 0)$. In basic order (x_1, x_2, x_3) , $c_{\mathcal{B}} = (1, 1, 1)$.

Restore the Original Objective by Substitution

The original objective is $f = x_1 + x_2 + x_3$. Substitute the remaining dictionary:

$$\begin{aligned} f &= (1 - \tfrac{1}{2}x_4) + (\tfrac{1}{2} + \tfrac{3}{4}x_4) + (\tfrac{1}{3} - \tfrac{1}{3}x_4) \\ &= \tfrac{11}{6} + (-\tfrac{1}{2} + \tfrac{3}{4} - \tfrac{1}{3})x_4 \\ &= \boxed{\tfrac{11}{6} - \tfrac{1}{12}x_4}. \end{aligned}$$

At $x_4 = 0$, the original cost is $11/6$. The coefficient $-1/12$ is its reduced cost: increasing x_4 can improve f .

For the tableau, write $-f - \frac{1}{12}x_4 = -\frac{11}{6}$:

$$R_0 = \left[-\frac{11}{6} \mid 0, 0, 0, -\frac{1}{12} \right].$$

Phase II Begins from a Feasible Original Basis

Basic	RHS	x_1	x_2	x_3	x_4
R_0	$-\frac{11}{6}$	0	0	0	$-\frac{1}{12}$
x_1	1	1	0	0	$\frac{1}{2}$
x_2	$\frac{1}{2}$	0	1	0	$-\frac{3}{4}$
x_3	$\frac{1}{3}$	0	0	1	$\frac{1}{3}$

Try the next pivot

Which variable can enter? Which basic variable limits the step? Compute the next point and original objective before continuing.

The following slides give the complete solution to this exercise.

Phase II: x_4 Enters and x_3 Leaves

Since $\bar{c}_4 = -1/12$, let $x_4 = \theta$. The basic values change as

$$\begin{aligned}x_1 &= 1 - \theta/2, \\x_2 &= 1/2 + 3\theta/4, \\x_3 &= 1/3 - \theta/3.\end{aligned}$$

$$\theta^* = \min \left\{ \frac{1}{1/2}, \frac{1/3}{1/3} \right\} = 1.$$

Basic x_3 leaves. Basic x_2 increases and gives no upper bound.

$$f_{\text{new}} = \frac{11}{6} - \frac{1}{12} = \frac{7}{4}.$$

Example 3.8: The Original Optimum Is 7/4

Basic	RHS	x_1	x_2	x_3	x_4
R_0	$-\frac{7}{4}$	0	0	$\frac{1}{4}$	0
x_1	$\frac{1}{2}$	1	0	$-\frac{3}{2}$	0
x_2	$\frac{5}{4}$	0	1	$\frac{9}{4}$	0
x_4	1	0	0	3	1

$$x^* = (1/2, 5/4, 0, 1), \qquad f(x^*) = 7/4.$$

The sole nonbasic reduced cost is $\bar{c}_3 = 1/4 > 0$. The feasible basis is optimal for the original minimization problem.

Check the original four equalities directly. The redundant equality still holds even though we no longer store it.

Phase I Ends, but an Artificial Variable Is Still Basic

$$\min f = -x_2 \quad x_1 + x_2 \leq 2 \quad x_1 + 2x_2 \geq 2 \quad x_1, x_2 \geq 0.$$

Add slack x_3 , surplus x_4 , and artificial a , all nonnegative. Minimize $w = a$.

$$\begin{aligned} x_3 &= 2 - x_1 - x_2, \\ a = w &= 2 - x_1 - 2x_2 + x_4. \end{aligned}$$

Start with basic $x_3 = a = 2$; nonbasic $x_1 = x_2 = x_4 = 0$.

Use Bland's rule with order x_1, x_2, x_3, x_4, a : x_1 enters. Both ratios are 2; x_3 leaves before a .

$$\begin{aligned} x_1 &= 2 - x_2 - x_3, \\ a = w &= -x_2 + x_3 + x_4. \end{aligned}$$

At $x_2 = x_3 = x_4 = 0$: $x_1 = 2$, $a = w = 0$. Since $w \geq 0$, **Phase I is optimal, but a is still basic.**

Replace the Artificial Variable Without Moving the Point

Before: basis (x_1, a) ; $x_2 = x_3 = x_4 = 0$, so the point is $(2, 0)$.

$$x_1 = 2 - x_2 - x_3 \quad a = -x_2 + x_3 + x_4.$$

Any of x_2, x_3, x_4 can replace a . Choose the smallest index, x_2 ; solve and substitute:

$$x_2 = x_3 + x_4 - a \quad x_1 = 2 - 2x_3 - x_4 + a.$$

Set nonbasic $a = 0$ and remove it:

$$x_1 = 2 - 2x_3 - x_4 \quad x_2 = x_3 + x_4.$$

After: basis (x_1, x_2) ; $x_3 = x_4 = 0$ gives the **same point** $(2, 0)$. This is a zero-step exchange: either coefficient sign works, with no ratio test.

After Phase I: Two Possible Cases

At $w = 0$, all artificial values are zero. Set **only artificial variables** to zero; keep the LP variables as symbols.

A constraint remains: keep it

$$a = -x_2 + x_3 + x_4 \xrightarrow{a=0} 0 = -x_2 + x_3 + x_4.$$

Solve for a variable that appears, then substitute into the other rows. Here: $x_2 = x_3 + x_4$.

The equation becomes $0 = 0$: remove it

$$x_7 = x_5 + x_6 \xrightarrow{x_5=x_6=x_7=0} 0 = 0.$$

This is the artificial row in Example 3.8. It imposes no restriction on the LP variables.

Selection convention: process basic artificials in index order; choose the smallest-index non-artificial variable that appears. Either coefficient sign works. Repeat until no artificial remains basic.

Example 3.8: Start with Fewer Artificial Variables

Recall Example 3.8 after reversing the second equality. Minimize $f = x_1 + x_2 + x_3$, subject to:

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 3, \\ -x_1 + 2x_2 + 6x_3 &= 2, \\ 4x_2 + 9x_3 &= 5, \\ 3x_3 + x_4 &= 1.\end{aligned}$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

x_4 appears only in the fourth row: $x_4 = 1 - 3x_3$. At $x_1 = x_2 = x_3 = 0$, $x_4 = 1$ **can already be basic**.

Add artificials only to the first three rows. Omit x_8 and the first pivot:

$$(x_5, x_6, x_7, x_4) = (3, 2, 5, 1) \quad w = x_5 + x_6 + x_7.$$

Phase I: The Complete Procedure

1. Construct a feasible auxiliary dictionary

Multiply rows with negative right-hand sides by -1 so $b \geq 0$. Keep usable basics; add needed artificials and substitute into $w = \sum_i a_i$.

2. Check the cost, then pivot if needed

At $w = 0$, go to step 3: $w \geq 0$ already proves optimality. If $w > 0$ and all reduced costs are nonnegative, the original LP is infeasible.

Otherwise select an entering variable with negative reduced cost; apply the stated pivot rule and ratio test. Pivot and repeat. Retained artificials are eligible too.

3. At zero cost, remove artificials

Fix nonbasic artificials at zero. For each basic artificial, use the two cases just shown: a zero-step exchange or deletion of a $0 = 0$ row. Then restore the original objective for Phase II.

Rules in our examples: Example 3.8 prescribes its pivots; the new example uses Bland. Break ratio ties by smallest basic-variable index.

Phase II: Restore and Optimize the Original Cost

1. **Keep the basis** obtained after removing artificial variables. It is feasible for the remaining independent original equalities.
2. **Keep the constraint rows** after removing artificial columns; discard the Phase I objective row.
3. **Restore the objective:** substitute the dictionary into $f = c^\top x$. Its constant is the current cost; the nonbasic coefficients are the reduced costs.
4. **Continue simplex** with an anticycling rule until an optimal basis or an improving feasible ray is obtained.

Two Phases Handle Every Standard-Form Outcome

What happens?	Where it is handled
The original LP is infeasible	Phase I optimum $w^* > 0$.
Feasible equalities are dependent	Remove redundant equality rows after Phase I.
Original objective is unbounded below	Phase II finds an improving feasible ray.
A finite optimum exists	Phase II terminates at an optimal basis.

An anticycling rule supplies the termination guarantee, including degenerate cases.

We now know how to start and finish. Practice the complete method in the **workshop**. Then **Simplex IV** interprets basis exchanges through the columns of the constraint matrix.

References and Paths for Independent Study

Practice the complete two-phase method

Construct Phase I, interpret its optimum, remove artificial variables, and restore the original objective.
Continue Phase II until you establish optimality or unboundedness.

Practice: Simplex problem-solving workshop: pivot rules, degenerate steps, two phases, and infeasibility.

Optional details: Why artificial variables can be removed. Review the worked dictionaries first.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §3.5.

Review: Simplex II: Cycling and Anticycling.

Continue: Simplex IV: Column Geometry and Efficiency.

Why Did One Equation Become $0 = 0$?

Look at the original equalities after reversing the sign of the second one:

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 3, \\ -x_1 + 2x_2 + 6x_3 &= 2.\end{aligned}$$

Add them. The x_1 terms cancel:

$$4x_2 + 9x_3 = 5.$$

This is an original constraint in Example 3.8. It follows from the two equalities above, so it is **redundant**.

Subtracting those two equalities from $4x_2 + 9x_3 = 5$ gives $0 = 0$. Removing this redundant equation loses no original feasible points.

If the Final Basis Already Uses Original Columns

Suppose the final auxiliary basis B consists entirely of columns of A . Keep its feasible basic values $B^{-1}b$.

Delete artificial columns. The original constraint rows remain

$$[B^{-1}b \mid B^{-1}A].$$

Using the original costs c , rebuild the objective information:

$$f_0 = c_{\mathcal{B}}^{\top} B^{-1}b, \quad \bar{c}^{\top} = c^{\top} - c_{\mathcal{B}}^{\top} B^{-1}A.$$

Here $c_{\mathcal{B}}$ lists costs in the same order as the columns of B . Continue simplex with this original objective.

Why Original Columns Can Replace the Artificial Columns

Suppose the final auxiliary basis contains $k < m$ original columns. Reorder the basis so these are $A_{B(1)}, \dots, A_{B(k)}$.

They are linearly independent because they are part of a basis.

If $\text{rank}(A) = m$, the columns of A span $\mathbb{R}^m$. Extend those k independent columns to m independent original columns:

$$\widetilde{B} = [A_{B(1)} \ \cdots \ A_{B(k)} \ A_{B(k+1)} \ \cdots \ A_{B(m)}].$$

The symbols $B(i)$ name column indices; B and $\widetilde{B}$ are basis matrices.

The Extended Basis Represents the Same Point

Every positive original component of x^* was already basic. Original variables outside those k columns were nonbasic and therefore zero.

The added original columns receive value zero. With all other nonbasic values zero,

$$\tilde{B}x_{\tilde{\mathcal{B}}}^* = b, \quad x_{\tilde{\mathcal{B}}}^* \geq 0.$$

Nonsingularity makes these the basic values $\tilde{B}^{-1}b$. Thus x^* is unchanged.

If $\text{rank}(A) < m$, no such m -column basis exists. We must identify and remove redundant equality constraints.

Inspect the Row of a Zero Artificial Variable

Let row ℓ belong to an artificial basic variable with value zero. Write $H = B^{-1}A$ and $h = B^{-1}b$; H_{ij} is row i , column j .

Value	x_1	...	x_j	...	x_n	Artificial column
h_i	H_{i1}	...	H_{ij}	...	H_{in}	0
$h_\ell = 0$	$H_{\ell 1}$	...	$H_{\ell j}$	...	$H_{\ell n}$	1
h_r	H_{r1}	...	H_{rj}	...	H_{rn}	0

Rows $i, r \neq \ell$ stand for the other constraint rows. The basic artificial column is the unit vector e_ℓ .

Search only the original columns: is some $H_{\ell j} \neq 0$? A nonzero entry permits a basis exchange; an entirely zero row reveals a redundancy.

Case 1: A Nonzero Entry Supplies a New Column

Keep the original basic columns first, so their positions are $1, \dots, k$ and the artificial row satisfies $\ell > k$. The vector e_i has a one in position i , zeros elsewhere.

$$B^{-1}A_{B(i)} = e_i \quad (i = 1, \dots, k).$$

Every linear combination of these vectors has zero ℓ th component.

$$H_{\ell j} = (B^{-1}A_j)_\ell \neq 0 \quad \implies \quad B^{-1}A_j \notin \text{span}\{e_1, \dots, e_k\}.$$

Because B is invertible, A_j is independent of the original columns already in the basis.

Pivot at Zero: Even a Negative Entry Works

Choose a nonzero original entry $p = H_{\ell j}$. Let R_i denote the entire i th tableau row.

$$R_\ell \leftarrow R_\ell / p, \quad R_i \leftarrow R_i - H_{ij} R_\ell \quad (i \neq \ell).$$

The entering column becomes e_ℓ . Original variable x_j enters; the artificial variable leaves.

Since $h_\ell = 0$, scaling this row and adding it to other rows changes no basic value. The same feasible point remains, with one fewer basic artificial variable.

The pivot may be negative: this operation exchanges basis columns at a zero step.

Case 2: Every Original Entry in the Row Is Zero

Suppose $H_{\ell j} = 0$ for every original column $j = 1, \dots, n$.

Value	All original columns	Artificial column
0	0 ... 0	1

Let $g^\top = e_\ell^\top B^{-1}$, the ℓ th row of the inverse basis matrix.

$$g \neq 0, \qquad g^\top A = e_\ell^\top B^{-1} A = 0^\top.$$

A row of an invertible matrix cannot be zero. Thus g gives a nontrivial linear dependence among the rows of A .

Why the Zero Row Can Be Removed

Phase I has reached zero, so an original feasible point x^* exists.

$$g^\top b = g^\top Ax^* = 0.$$

The same combination of right-hand sides is therefore also zero.

$$g^\top Ax = g^\top b \iff 0 = 0.$$

After artificial variables are set to zero, this transformed tableau row imposes no restriction. Delete that redundant row and its artificial basic variable.

Repeat: pivot out a zero artificial variable when possible; otherwise delete its redundant row. Finish with an original-variable basis for the remaining independent constraints.